

UCCD3094 Principles of Deep Learning

Group Assignment

Cross-Domain Generalisation: Evaluating CNN-Transformer Architecture and Augmentation Scaling

Research-based ☒ Application-based ☐

1. INTRODUCTION

Plant diseases cause losses of up to 40% of global food production annually [1], making automated identification from leaf images an important research area. While deep learning models reach 98% accuracy on clean controlled datasets like PlantVillage, performance drops significantly when tested on field datasets like PlantDoc [2]. This drop shows that good results on a clean dataset do not necessarily translate over to real-world use. The motivation is to address the cross-dataset gap by applying increasing factors of data augmentations on three deep learning architectures, as well as developing a hybrid architecture that combines the strengths of Convolutional Neural Networks (CNN) with Vision Transformers (ViTs).

CNNs excel at local lesion texture, but may miss global context, while ViTs model long-range dependencies via self-attention but struggle with fine-grained tasks under data constraints. Hybrid CNN–Transformer architectures have been proposed to combine both strengths [3, 4], motivating the approach taken here. A hybrid CNN–Transformer model was created by fusing EfficientNetV2 and Swin Transformer for both local feature extraction and global context modelling. Rather than the conventional direction of training on a clean dataset, PlantVillage, and testing on real-world dataset, PlantDoc, this project reversed the protocol. The models were trained on PlantDoc and evaluated on PlantVillage via a zero-shot approach. Removing environmental noise during testing allowed us to determine whether the model can truly separate visual features from the background clutter.

2. RELATED WORK

Current research on the “lab-to-field” gap highlights a severe performance drop when models transition from controlled datasets like PlantVillage to high-variance PlantDoc dataset. While models trained on PlantVillage manage to achieve up to 99% accuracy, their performance drops to 15% to 40% on PlantDoc due to the extreme domain shift, such as illumination shifts and occlusions [2]. Specifically, VGG16 achieved 15.08% on uncropped PlantDoc images, while

InceptionV3 and Inception ResNet V2 reached 30.78% and 39.87% on the cropped dataset, respectively. Studies suggest the drop is caused by “spurious correlations”, where models learn environmental cues (soil, lighting) instead of leaf pathology [5, 6].

Architectural approaches to mitigate this gap focus on balancing local and global features. CNNs such as EfficientNetV2 offer strong spatial inductive biases for fine morphological details but remain susceptible to background noise [7, 8]. In contrast, hierarchical models such as Swin Transformer utilise self-attention to model global patterns and reduce background inference, yet they lack the local inductive biases required for noisy pathological details [9][10]. Recent hybrid architectures, such as ConvTransNet-S, attempt to integrate these complementary strengths to improve domain generalisation [11].

To further bridge the domain gap, data augmentation techniques such as Enhanced-RICAP and colour jitter were used to simulate field noise [12]. However, excessive augmentation risks “over-augmentation”, where the models fit to synthetic noise rather than biological features [13, 14]. This study addressed this by proposing a scalable 1–12x data augmentation pipeline utilising CLAHE to preserve local lesions under distortion, coupled with randomised weighted sampling to maintain class balance [15, 16].

3. SYSTEM DESIGN

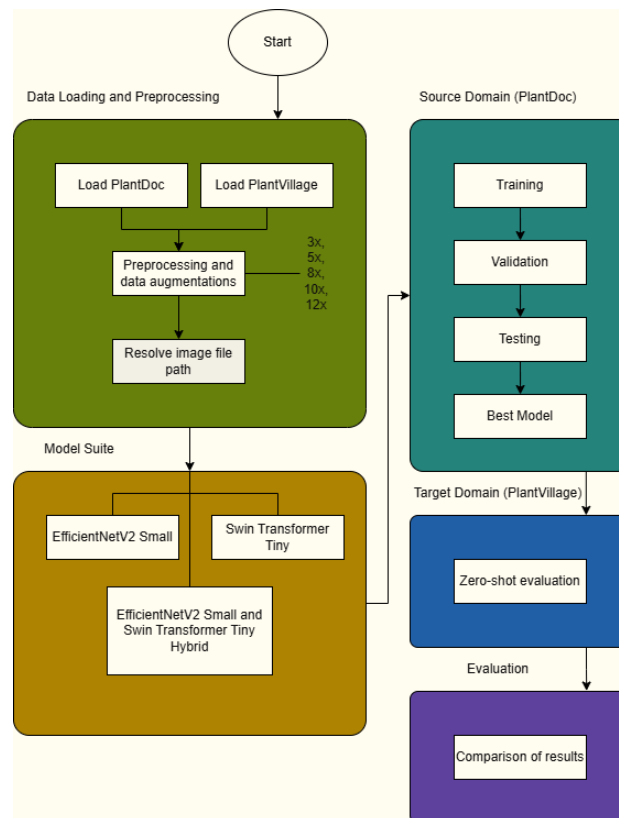


Figure 3.1 Overview of Pipeline

Data Loading and Preprocessing

This phase focused on preparing PlantDoc and PlantVillage for cross-domain experiments. Due to the high complexity of PlantDoc, aggressive augmentations were applied to it, such as CLAHE, geometric transformations, and random erasing. In contrast, standard augmentations, such as resizing and converting to Pytorch’s tensor format, were applied to PlantVillage. The inclusion of CLAHE preprocessing was to enhance local details in plant images and adjust shadows, potentially aiding the model’s ability to identify subtle details. To address cross-domain transferability, data augmentations were applied to PlantDoc at various rates (x1, x3, x5, x8, x10, and x12). Random weighted sampling was utilised across both datasets to mitigate inherent class imbalances.

Model Suite

Three architectures were benchmarked: EfficientNetV2-S, Swin-S variant, and a hybrid CNN-vision transformer fusing both EfficientNetV2-S and Swin-S. By benchmarking EfficientNetV2, which focuses on local spatial features [7], against Swin, which focuses on global contextual relationships [9], and a hybrid model, a combination of two revolutionary models, we could determine how each model handles domain shift and the effect of the varying augmentation factors. Each model was initialised with the same weight, IMAGENET1K-V1, to ensure fair comparison. To prevent the aggressive overriding of these pre-trained features while allowing for domain-specific adaptation, all backbones are unfrozen and trained with low learning rates.

Source Domain (PlantDoc)

Each model was trained under the same hyperparameter set; differential learning rate was implemented with a learning rate of 1×10^{-5} for the backbones and 1×10^{-4} for the classification heads. AdamW with L2 initialisation was used, followed by a linear warmup-cosine annealing combination, providing stable training. A checkpointing and early stopping system was implemented, halting training if the model did not improve for 10 consecutive epochs to prevent training overfitted models. Furthermore, considering that it was a multiclass classification problem, the cross-entropy loss function was used. Finally, for more efficient training, the mixed precision technique was used.

Target Domain (PlantVillage)

The best checkpoint from each model was chosen and subsequently evaluated on PlantVillage using a zero-shot approach. By bypassing fine-tuning on the target domain, we isolate the

impact of the source-domain augmentation factors on the model’s ability to extract pathological features.

Evaluation

Each model was evaluated and studied against each other to determine the extent and magnitude of augmentation factors on their cross-domain generalisability. The models were evaluated using accuracy, F1-score, and confusion matrices.

4. EXPERIMENT & EVALUTION

Zero-shot Accuracy vs Augmentations

We used zero-shot accuracy as the primary metric to demonstrate the model’s generalisability. This approach highlights performance gains achieved through increased augmentation. Figure 4.1 demonstrates the effect of augmentation levels on models’ cross-domain generalisability.

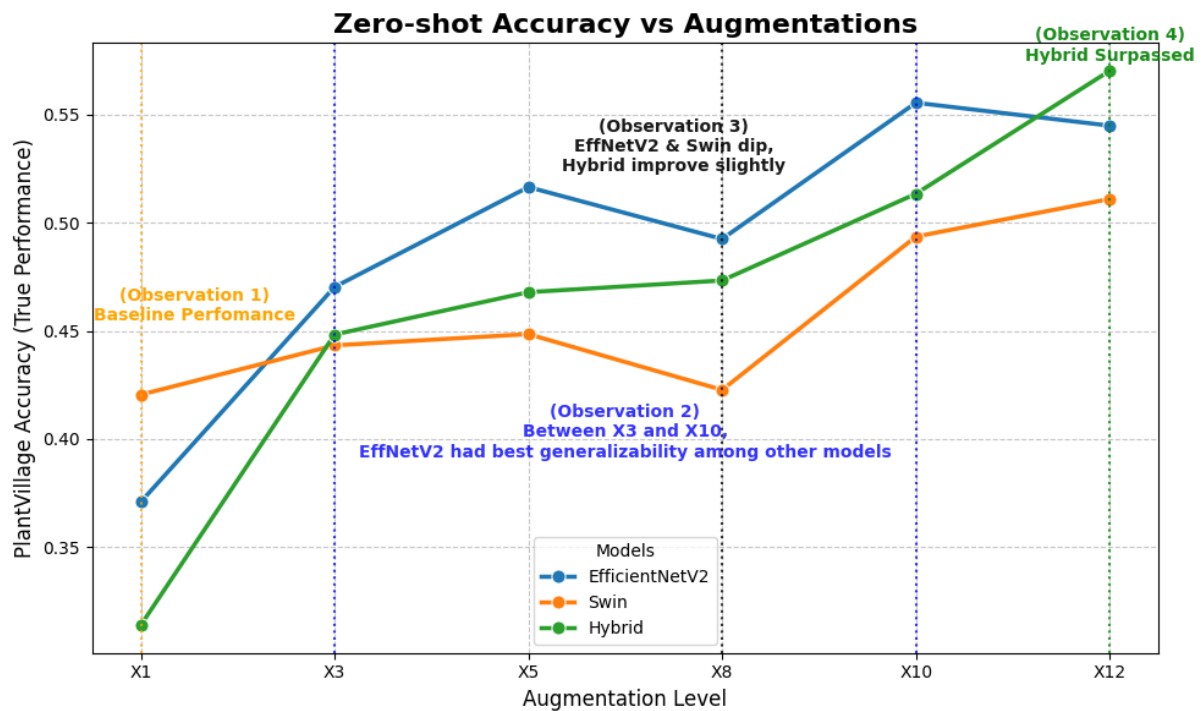


Figure 4.1 Zero-shot Accuracy vs Augmentations with Observation Label

Based on the figure, we could make three observations:

Observation 1: At x1 (factor of 1), the Swin model (0.42) shows a strong baseline performance in zero-shot task, followed by EfficientNetV2 (0.37) and Hybrid (0.31). This advantage arises because the transformer’s global attention enables Swin to separate background noise from leaf structures using image-wide context, allowing it to focus directly on leaf pathology with fewer augmentation and samples.

Observation 2: Between x3 and x10, EfficientNetV2 consistently topped other models with a gradual increment in zero-shot accuracy, peaking at 0.56. However, at x12, the model accuracy fell to 0.54. Despite the dip, the model still demonstrated strong generalisability and stable performance at moderate augmentation numbers. The EfficientNetV2 architecture is optimised for spatial information, so the high augmentation level (x12) may have caused overly aggressive distortions on the samples. Consequently, this disrupts the local spatial rules that the EfficientNetV2 relied on, causing it to lose cues of the diseases amongst the noises.

Observation 3: At x8, EfficientNetV2 (-2.40%) and Swin (-2.58%) models show significant dip while Hybrid (+0.54%) show slight improvement compared to x5. This indicates that at x8, the augmentation introduced distortions that disrupt the EfficientNetV2 and Swin models due to insufficient diversity of data representation for better generalisation. In contrast, Hybrid model was able to overcome this distortion due to CNN and Transformer capability to extract cleaner global features. Importantly, all three model zero-shot performance recover and improves after augmentation x8. This suggest that increasing augmentation level can still lead to generalisation gains

Observation 4: The Hybrid model exhibits a steady performance increase across augmentation levels, at x12, the Hybrid (+5.69%) model showed a significant performance gain, increasing from 0.51 to 0.57. The model surpassed the peak performance of EfficientNetV2 (x10 and x12 model), possibly indicating that Hybrid architecture scales most effectively with higher data augmentations, given its high parameter count. This suggests that the fusion of CNN-based spatial inductive biases with Transformer-based global context creates a more robust feature representation. High-intensity augmentation could have acted as a regularisation mechanism, forcing the model to ignore non-leaf noise and focus on domain-invariant features (leaf morphology).

Based on these observations and as illustrated in Figure 4.2, immediate intuition suggests that higher validation scores and lower overfitting lead to better models; however, these results demonstrate that source-domain success (PlantDoc) is a poor predictor of zero-shot cross-domain transferability (PlantVillage).

Overfitting leads to better Zero-shot Transfer

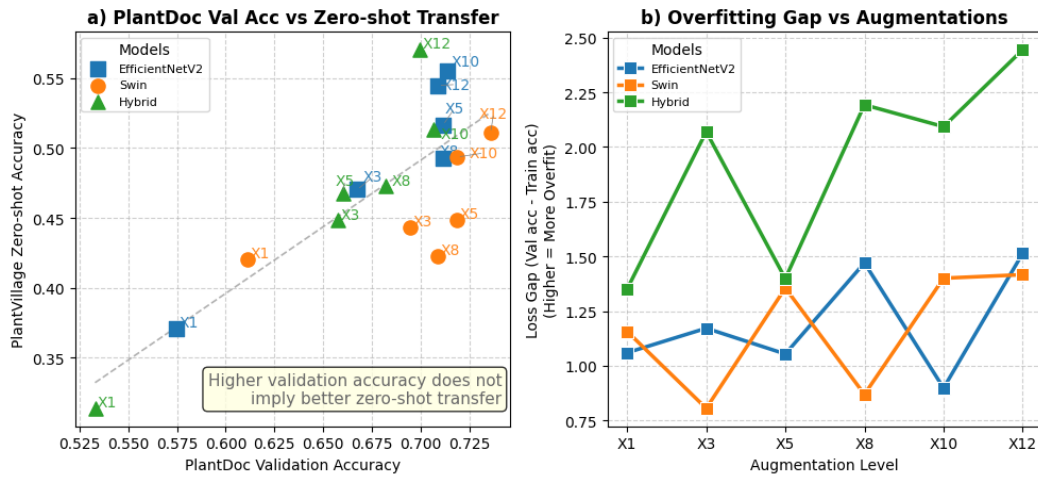


Figure 4.2 (a) PlantDoc Val vs Zero-shot Accuracy; (b) Overfitting Gap vs Augmentation

Validation Accuracy (Fig. 4.2a): At the x12 augmentation level, the Swin model achieved the highest validation accuracy (0.73) but the lowest zero-shot performance (0.51). In contrast, the Hybrid model shows a lower validation score (0.70) but achieved the highest zero-shot accuracy (0.58). This indicates overfitting to domain-specific artefacts that weren't present in PlantVillage. The Swin model's global attention design, where they learn relationships between every image patch across the entire frame, may have prioritised environmental noise over pathological features due to the increased appearance of noise (watermark, hands, or banners) from the larger augmentation factor. Consequently, the Swin model learns to identify the context of the images, which were heavily influenced by noise, rather than the diseases.

Overfitting Gap (Fig. 4.2b): The Hybrid model consistently exhibited the largest loss gap (peaking at 2.48 at x12). Based on intuition, this could have indicated poor zero-shot performance; however, at the highest augmentation factor (x12), the Hybrid model outperformed other models. The combination of CNN's design for local information and Swin Transformer's global attention capability could have resulted in a model capable of capturing disease-specific information (local) while filtering noises (global). Thus, when applied to PlantVillage, it was able to extract the necessary plant disease features.

Table 4.1 shows the numerical output of the performances of the models; it shows that the performance of all three models increases along with the augmentation scaling. Additionally, aside from the improving performances, the close alignment between F1 Macro and F1 Weighted scores (0.01 - 0.04) suggested that the implementation of random weighted sampling successfully mitigated class imbalance.

Table 4.1: Model Performance by Augmentation Scale

Augmentation	Model	Accuracy	F1 Macro	F1 Weighted
X1	EfficientNetV2	0.3709	0.3792	0.3478
	Swin	0.4205	0.4245	0.3841
	Hybrid	0.3139	0.3206	0.2978
X3	EfficientNetV2	0.4703	0.4611	0.4582
	Swin	0.4582	0.4158	0.4233
	Hybrid	0.4483	0.4117	0.4329
X5	EfficientNetV2	0.5165	0.4884	0.4858
	Swin	0.4485	0.4276	0.4222
	Hybrid	0.4679	0.4337	0.4340
X8	EfficientNetV2	0.4925	0.4841	0.4827
	Swin	0.4227	0.4018	0.4002
	Hybrid	0.4733	0.4647	0.4707
X10	EfficientNetV2	0.5555	0.5204	0.5205
	Swin	0.4936	0.4504	0.4811
	Hybrid	0.5134	0.4791	0.5047
X12	EfficientNetV2	0.5335	0.5107	0.5245
	Swin	0.5105	0.4954	0.5019
	Hybrid	0.5645	0.5100	0.5531

5. CONCLUSION

This study evaluated the cross-dataset robustness of EfficientNetV2, Swin Transformer, and a hybrid CNN–Transformer using a “dirty-to-clean” zero-shot protocol. Results demonstrate that augmentation scaling plays a critical role in generalisation. We found that pure CNN models perform better with a lower augmentation factor, given that the data is noisy, to avoid unfavourable disruption to its local inductive biases design. Swin model, however, may struggle overall from noisy data due to its inherent tendency to look at images at a global level, amplifying the impact of noisy data and consequently causing overfitting. Finally, CNN-Transformer hybrid models can perform well on noisy data; however, larger augmentation factors must be applied to unlock their generalisation potential. Future work should investigate the point in which performance starts to converge across more varied models or other specialised adaptation techniques.

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